London School of Economics: FM445 Portfolio Management

Portfolio Management in a Low-Growth World: Repricing Profitability under Secular Stagnation

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1 Introduction

In the aftermath of the devastating 2008 financial crisis, macroeconomists revived the debate around secular stagnation, a theory of persistently low aggregate demand first popularised by Alvin Hansen following the Great Depression (Summers 2015, 61). This theory predicted a decline in aggregate growth and interest rates, such that monetary policy would be ineffective at stimulating a flagging economy. While recent phenomena like the post-pandemic inflation and AI boom have temporarily dispelled fears of permanent slowdown, the structural conditions underpinning secular stagnation remain in force (Benigno et al. 2024). With the possibility of stagnation's return, it is important to understand its potential impact on portfolio management, so that managers can adjust their investments to possible structural changes in asset pricing.

Secular stagnation is defined by a persistently low natural rate of interest that causes the Zero Lower Bound (ZLB) of the central bank policy rate to be binding, alongside a secondary condition of low potential output growth. The natural rate of interest is the interest rate at which full employment generates a perfect absorption of savings by investment (Rachel and Summers 2019, 2–3). While economic output is typically operationalised through gross domestic product (GDP), potential output is a theoretical concept delineating the level of "output consistent with steady nonaccelerating inflation" (Gordon 2015, 54). The complementary aspects of secular stagnation, the low natural rate of interest and low potential output growth, primarily spring from two 2015 articles on secular stagnation. In one, secular stagnation is defined by the happenstance of reduced investment demand and increased savings, driving down the natural rate of interest and leading to a binding ZLB (Summers 2015, 62–64). In the other secular stagnation is defined by low potential economic growth due to structural factors such as sluggish productivity growth, declining labour force participation, and slowing population growth (Gordon 2015, 58).

Consequences of secular stagnation on the financial system have been noted by many economists, with particular focus on asset bubbles (Summers 2015, 64), but there has been limited research of the direct effect of secular stagnation on investment strategies. The literature has focused on the impact of individual stagnation characteristics on asset prices. For instance, Kroen et al. (2021) find that changes in interest rates near the ZLB affect the market price of industry leaders more than other firms. Another paper finds that expectations of low future growth lead both to depression in the real interest rate and lower asset prices (Kung and Schmid 2015). Israel et al. (2021, 3) find that the "low-interest-rate environment" does not explain underperformance of value investing.

We examine three factor portfolios commonly used in the asset pricing literature. The first is value, which sorts on the book-to-market ratio with a HML construction following Fama and French (1993). For our controls we use momentum and quality. The momentum portfolio sorts on stock cumulative returns

from one year ago to two months ago (UMD), as per Jegadeesh and Titman (1993) and Carhart (1997). The quality portfolio as constructed by Asness et al. (2019) sorts on a composite of variables including profitability, safety, and growth.

By investigating the channels for secular stagnation's effect on portfolio performance, we contribute to knowledge of portfolio performance under persistently adverse macroeconomic conditions. While previous papers on the subject focus on strategy returns and the relation between asset prices and stagnation, we test a particular mechanism through which stagnation acts on investment strategies.

1.1 Hypotheses and mechanisms

We hypothesize that secular stagnation affects factor strategies' loadings on profitability such that the value loading on profitability declines due to a change in the portfolio's tilt on profitability, while the quality and momentum loadings on profitability are not affected in this way. The quality (QMJ) loading on profitability (RMW) merits some explanation, as both the quality portfolio and the RMW factor portfolio sort on profits but are constructed differently. QMJ uses multiple measures of profitability, including gross profit over assets, gross margin, and return on assets, while RMW uses operating profit over book equity (Fama and French 2015).

The change in profitability loadings occurs because the characteristics of stagnation, low interest rates and low growth, increase profitability's salience for market pricing. One channel for this change is discount rate compression. Since firm price is the present value of market-expected cash flow, the low natural rate of interest shifts value to far-off cash flows. In other words, distant cash flows make up disproportionately more of present value. Firms with durable (long-term) profits are more highly valued due to the decline in interest rates, driving up these firms' market price.

The second channel for this change is growth scarcity. Under secular stagnation, aggregate growth is weak and predicted to be so in the long term, causing the market to pay more for firms with durable profitability. Firms predicted to have durable profits are more scarce under low aggregate growth conditions, increasing their price on the market. Novy-Marx (2013) finds that current profitability is a strong predictor of future profitability and thus profitable firms have longer cash-flow durations.

The lower loading of value on RMW follows from these channels. Firms expected to have more value in distant cash flows are bid up in the market, as they stand to gain most from secular stagnation interest rates. As their market prices increase, these firms' book-to-market ratios decline and the firms move from the long leg of the value portfolio to the short leg. Firms in the high B/M long leg are increasingly those without durable profits. Growth scarcity lowers profitable firms' B/M, reinforcing the drive from the long to the

short leg of the value portfolio. In the portfolio, the migration of profitable firms from the long to the short leg causes the value portfolio to tilt away from operating profitability over book equity, the profitability factor. This more negative tilt on OP/BE causes the value portfolio to covary more negatively with the RMW factor, a negative loading.

The quality portfolio (QMJ) includes multiple measures of profitability in its sorting mechanism, so it should have a positive tilt toward OP/BE, another measure of profitability. QMJ does not sort on valuation, thus its tilt will be unaffected by a repricing of profitability, leading to no loading change. Since momentum is entirely past-based and does not sort on expected profit durability, it should exhibit no OP/BE tilt and its loading should be unaffected by stagnation.

Given that value's loading on profitability should decrease while quality's loading should remain positive, we expect their returns to become more negatively correlated under stagnation. We cannot make the same prediction for value and momentum, as momentum should not have any profitability loading to begin with, and thus a change in value's loading will not affect their returns correlation through the profitability channel.

Ultimately we find support for all of our hypotheses with the exception of quality. Value changes in the predicted direction, with tilt declining from -0.27 to -0.39 under stagnation and a reversal in the sign of its loading on profitability. Momentum's lack of change lends support to our proposed channel, but quality does exhibit a change in tilt and loading that we do not predict, though its comovement with profitability remains unchanged.

2 Data

This paper uses monthly stock data from 1989 to 2024 provided by the Center for Research in Security Prices and annual accounting data from Compustat, accessed through WRDS. The first month of analysis is July 1990, as the book-to-market sort requires December market equity from the preceding fiscal year. Our factors derive from the Fama-French five-factor model, which includes market excess return, SMB, HML, RMW, and CMA (Fama and French 2015). RMW sorts on operating profitability and CMA sorts on asset growth. The one-month Treasury bill rate is used as the risk-free rate, pulled from Kenneth French's data library (French 2026). Financial firms are retained in the data, following the Fama–French factor portfolios.

This study analyses the performance of portfolios using common shares (denoted by CRSP share codes 10 and 11) listed on the NYSE, AMEX, or NASDAQ. Two filters are used to exclude potentially problematic data: a minimum price and a monthly winsorization. A \$5 minimum price is a standard screen that excludes microcap stocks to rule out spurious volatility and negative autocorrelation, following Jegadeesh and Titman (2001). A monthly cross-sectional winsorization at the 1st and 99th percentiles is used to limit the impact

of extreme outliers and data errors on regression estimates. In total, we analyse 1,585,887 monthly stock observations across 17,262 unique stocks.

Our factor portfolios of value (HML), quality (QMJ) and momentum (UMD) rely on characteristics in the universe of stocks that we calculate. Value sorts on the book-to-market ratio of stocks, here calculated by book equity from the prior fiscal year over market equity calculated in December. Quality sorts on the Asness, Frazzini, and Pedersen composite, and momentum sorts on the cumulative return of a stock from month t-12 to month t-2. In order to avoid look-ahead bias in our data, portfolios formed in year t are based on accounting data available in year t-1.

The natural rate of interest and potential output growth cannot be directly observed, so they must be estimated. Most estimates put the natural rate of interest as low or negative from 2009 to 2021, but these contemporary estimation tools remain highly uncertain (Benigno et al. 2024, 17–22). Potential output is also a difficult concept to estimate. Measurement of the "output gap", the gap between realised output and potential output is subject to high levels of uncertainty (Orphanides 2001, 970). This paper takes the simple approach of operationalising potential output growth through GDP growth. There are various structural factors that have put downward pressure on GDP growth since even before 2009, such as the aging American population (Maestas et al. 2023, 317–18). This low GDP growth, combined with the low estimated natural rate of interest over the 2009-2021 timeframe make this our "secular stagnation" period candidate.

The macroeconomic regimes chosen for this analysis are the normal regime (July 1990 - November 2007), secular stagnation (July 2009 - November 2021), and a rising-rate period (July 2022 - December 2024). December 2007 through June 2009 are excluded as they are the NBER-dated recession period, during which conditions are characterised neither by normal conditions nor secular stagnation. The stagnation period is defined by the estimates of low-negative natural rate of interest found in Benigno et al. (2024). The normal period comprises 209 months, the stagnation period comprises 149 months, and the rising-rate period comprises 30 months. The final period is short and follows an excluded policy-transition window; it is used primarily as an out-of-sample robustness check.

After positive book equity and valid December and June market equity are required, the total number of firm-years in the sample is 110,733. Table 1 shows the three sorting characteristics and two measures of profitability. Since Book-to-market is calculated annually, it has an order of magnitude fewer observations than the other characteristics. Book-to-market has a mean of 0.63 and median of 0.52. Gross profitability has a mean of 0.28 and a median of 0.24. Momentum has a mean of 0.12 and a median of 0.06; its dispersion reflects heavy tails in cumulative returns over its eleven-month period. OP/BE and GP/AT have a rank correlation of 0.32 in the normal regime and 0.41 under stagnation.

Table 1: Distribution of sorting characteristics

Characteristic	Unit	N	Mean	SD	p10	Median	p90
A. Sorting variables							
Book-to-market (B/M)	firm-years	110,733	0.627	0.529	0.153	0.522	1.175
Quality score (QMJ composite)	firm-months	1,244,618	0.000	1.000	-1.385	0.000	1.385
Momentum (12-2 return)	firm-months	1,379,957	0.115	0.485	-0.376	0.064	0.606
B. Profitability measures							
Gross profitability (GP/AT)	firm-years	110,377	0.283	0.265	0.032	0.243	0.635
Operating profitability (OP/BE)	firm-years	110,379	0.206	0.426	-0.020	0.218	0.458

Table 2: Quality portfolio score coverage

	Share of firm-months
Profitability composite	0.852
Growth composite	0.573
Safety composite	0.783
Quality score (2 of 3 required)	0.785
Firm-months in the panel	1,585,887

Table 3: GP/AT vs OP/BE cross-sectional rank correlation

	Months	Mean	SD	Range
Normal	209	0.315	0.050	[0.224, 0.411]
Stagnation	149	0.410	0.030	[0.347, 0.502]

Table 4 reports annualised factor moments in the normal period, stagnation period, and rising-rates period. The factor moments chosen are mean, volatility, and Sharpe ratio. The premium earned by value declines from 2.4% in the normal period to −5.4% under stagnation. The premium earned by quality increases from 2.6% to 3.4% in the same timeframe, while momentum falls from 8.8% to 2.9%. Sharpe ratios move in the same directions, and volatility is comparable across regimes. The profitability premium does not change significantly, decreasing from 4.50% to 2.67% under secular stagnation with an insignificant t-statistic of −0.54, while its volatility drops from 10.9% to 6.3%. The rising-rate period is short, only 30 months, and is used exclusively as an out-of-sample check.

Table 4: Annualised factor moments by regime, July 1990–December 2024

	Normal			Stagnation			Rising		
	Mean	Vol	SR	Mean	Vol	SR	Mean	Vol	SR
Value (HML)	2.44	12.4	0.20	-5.40	11.8	-0.46	-9.43	16.5	-0.57
Momentum (UMD)	8.80	18.8	0.47	2.89	16.3	0.18	0.18	22.0	0.01
Quality (QMJ)	2.56	6.4	0.40	3.37	7.8	0.43	6.71	7.6	0.88
RMW (benchmark)	4.50	10.9	0.41	2.67	6.3	0.43	5.17	8.4	0.62

Table 5 shows the change in strategy annualised performance across regimes. The change in mean reflects the declines for value and momentum and increase in quality spread reported above, but the t-statistics for each fall short of 1.96, making these changes insignificant at the 95% confidence level for a normal distribution. With 358 monthly observations in the sample (209 normal regime months and 149 stagnation months), the t-statistic follows an approximately standard normal distribution.

Table 5: Performance change across regimes

	Δ annualised mean (pp)	HAC SE	t
Value (HML)	−7.85	5.63	−1.39
Momentum (UMD)	−5.91	6.21	−0.95
Quality (QMJ)	0.81	3.04	0.27

The value spread is comparable to that of the Fama-French value factor, and the HML series has a monthly correlation of 0.90 to the Fama-French HML over the pooled regime sample. These correspondences suggest that the following results are not outcomes of construction error. The gap between the Fama-French and the constructed HML is likely due to methodological differences, such as winsorization and the $5 filter.

3 Empirical Analysis

Portfolios are formed according to the following methodologies. The value portfolio is formed by a June ranking of firms on book-to-market ratio using NYSE breakpoints. The cross-section is split into quintiles based on the book-to-market ranking, and the value portfolio is long Q5 (the highest book-to-market quintile) and short Q1 (the lowest). NYSE breakpoints are computed from NYSE stocks only. Rebalancing occurs every June.

The quality portfolio is rebalanced monthly on the sixteen variables specified in Asness et al., using a sort on 30/70 NYSE breakpoints. This sort is conditional on size then quality in order to make the portfolio neutral with respect to firm size. Momentum rebalances monthly on cumulative returns from month t-12 to t-2 into NYSE quintiles, long high return and short low return quintiles. This portfolio is computed monthly, and the month-long gap between the end of the ranking window (t-2) and the holding month (t) avoids short-horizon reversal in sorting stocks.

The safety composite includes a betting-against-beta measure, tilting QMJ toward low-beta stocks. We depart from Asness et al.'s construction of the quality portfolio that uses daily returns to estimate β; our estimation of β uses a 60-month rolling OLS regression of each stock's monthly excess return on the market excess return. The $5 filter reduces the effect of nonsynchronous trading bias in beta estimates by excluding the low-price securities most affected by it (Scholes and Williams 1977; Frazzini and Pedersen 2014).

NYSE breakpoints are used for the portfolios to match the Fama-French convention and because AMEX and NASDAQ firms are more numerous and smaller. If breakpoints were generated from firms listed across all exchanges, boundaries would be determined by the microcap population that are negligible in market value, and cutoffs would drift over time as the composition of the listings changes, undesirable given that we are comparing portfolio behaviour across macroeconomic regimes. Using the NYSE breakpoints means that the quintiles are not equal in firm count, as non-NYSE firms are younger and tilt toward lower book-to-market ratios.

To evaluate performance of the chosen portfolios, we do two sets of regressions. First we regress each strategy within the normal-rate and secular stagnation regimes, then we regress each strategy on the regimes pooled together. The sub-regime regressions generate estimates of strategies on different factors, while the pooled regressions generate the standard error on the difference between the regime loadings. We regress on the market excess return (as per the CAPM) and on Fama and French's five factors. The value strategy differs from the other two in excluding the Fama-French HML factor, as we have established a high correlation between our constructed portfolio and the factor (0.9). Including the Fama-French HML would catch nearly all of our value strategy's variation, leaving no interpretability for the other regressors. On the other hand, we include RMW for our quality portfolio's regression, as we are particularly interested in the loading of quality on profitability over macroeconomic regimes.

In the combined regimes regressions we use a dummy, D, to represent inclusion of the secular stagnation regime, such that D=1 from July 2009 to November 2021, and D=0 otherwise. The dummy's coefficient measures the change in risk-adjusted performance across macroeconomic conditions, and the coefficient on RMWxD measures change in profitability exposure across regimes.

We use Newey-West standard errors with six lags, as monthly factor returns are serially correlated and heteroskedastic.

We compute the first principal component of the strategy returns series in order to determine the dimensionality of the factor space. The first principal component's share of total variance will tell us how much the total variance of the three strategies' returns are driven by one force without specifying what explains the variance. A first principal component approaching 1 reports a single common force explaining the variance of returns among all strategies.

4 Results

We find that value's tilt on OP/BE does indeed decrease under stagnation, decreasing from -0.27 to -0.39. Momentum stays near-zero with an increase that is not statistically significant. Since momentum does not

experience a change in its OP/BE tilt, the change in the other strategies' tilts cannot be ascribed to a generic post-2008 change that has no relationship to profitability. Quality's tilt slightly increases with statistical significance, rejecting our hypothesis that quality's tilt would be unaffected. Unlike momentum, quality does have an OP/BE tilt (of similar magnitude to that of value). Quality's tilt change, on the other hand, is smaller in magnitude than that of value, suggesting that whatever about the stagnation period causes quality to increase its OP/BE tilt is a weaker effect than the valuation-sort driving value's tilt.

In figure 1, we can see that the OP/BE spread in the value portfolio is continually wider in the shaded stagnation period than in the normal period, while GP/AT's spread remains roughly the same from the end of the normal period through stagnation. Since months December 2007 through June 2009 are excluded from our periodization, they appear as a gap in the figure.

Table 6: Portfolio tilts on OP/BE across regimes

	Normal	Stagnation	Δ	t
Value (HML)	-0.271	-0.390	-0.118	(-11.19)
Momentum (UMD)	0.025	0.052	0.027	(1.11)
Quality (QMJ)	0.230	0.292	0.062	(3.16)

Newey-West standard errors are used for t-statistics.

Figure 1: Profitability tilt of value

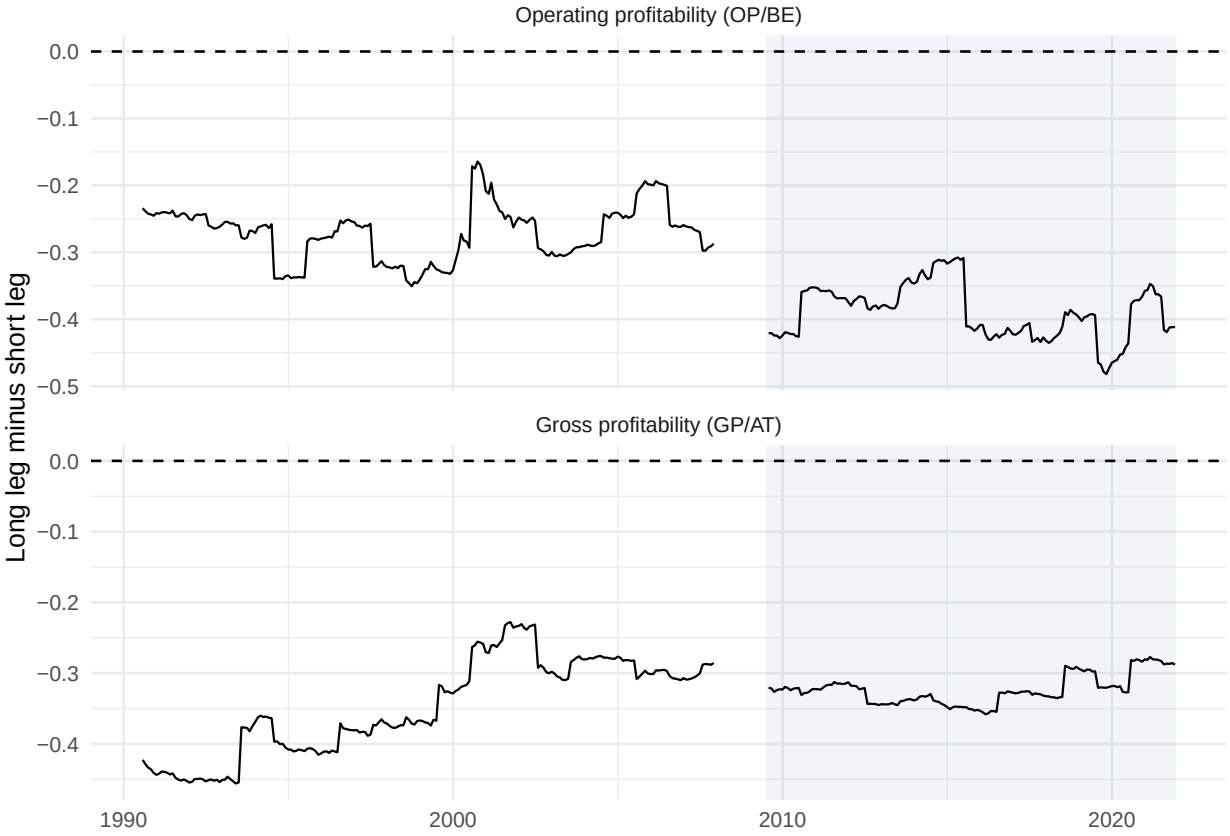


Table 7 demonstrates how firms with high operating profitability drift from the long to the short leg of the value portfolio under stagnation. The long leg's OP/BE decreases from 0.18 to 0.16, while the short leg's OP/BE increases from 0.45 to 0.54. Furthermore, the quality share of each leg does not change much. 72% of shares by market value in the short leg, Q1, are in the long leg of the QMJ portfolio under the normal regime, while 73% are in the long leg of QMJ under stagnation. Similarly, 5% of those in Q5 of the value portfolio are in quality's long leg under the normal regime, while 6% are in the long leg under stagnation. This means that the migration of firms in the value portfolio under stagnation is attributable to operating profitability in particular rather than general measures of quality.

Table 7: Profitability of value portfolio's legs across regimes

	OP/BE		GP/AT		Quality-leg share	
	Normal	Stag.	Normal	Stag.	Normal	Stag.
Value (Q5)	0.180	0.155	0.142	0.089	0.050	0.058
Growth (Q1)	0.451	0.544	0.489	0.412	0.723	0.726
Value − Growth	−0.271	−0.390	−0.347	−0.323	−0.673	−0.667

The factor loadings across macroeconomic regimes change in the following ways, as seen in table 8. Value's loading on profitability decreases by 0.59 under stagnation, while quality's loading increases by 0.18. As we predict, momentum's loading does not significantly change, with its measured loading change being negative but with a t-statistic of -0.74. That quality's loading significantly increases contradicts our hypothesis that stagnation would not impact quality's loading on the profitability factor. Importantly, the changes in alpha for all three strategies across regimes are insignificant at the 5% level, indicating that the difference in returns across regimes reflects changes in factor exposure rather than changes in unexplained performance. Momentum earns a significant positive alpha in the normal regime, and in line with the findings in Asness et al. (2019), quality earns a significant positive alpha in both regimes.

Table 8: Key coefficient shifts across regimes

	α	$\Delta\alpha$	β_{RMW}	$\Delta\beta_{\text{RMW}}$
Value (HML)	-0.003	-0.002	0.349	-0.592
	(-1.44)	(-0.67)	(3.06)	(-2.30)
Momentum (UMD)	0.009	-0.005	0.148	-0.230
	(2.05)	(-1.02)	(0.54)	(-0.74)
Quality (QMJ)	0.004	0.000	0.444	0.175
	(4.37)	(0.12)	(12.18)	(2.27)

α and β_{RMW} are normal-regime baselines, while the Δ terms are the coefficients on D and RMWxD. Newey-West standard errors are used for t-statistics.

The partial correlations test whether each strategy's loading change reflects closer comovement (table 9). We observed value's loading reversing sign from 0.35 to -0.24, which cannot be produced by a change in relative volatility. Relative volatility scales the loading of a strategy on a factor, but will not change its sign, as volatility is always positive. This loading reversal must then derive from comovement, confirmed by the flipped sign in its partial correlation with RMW, which goes from 0.32 to -0.15.

Further confirming our hypothesis, momentum's partial correlation remains near zero in both macroeconomic environments. Quality demonstrates no change in comovement; its partial correlation with RMW moves only from 0.68 to 0.66. Quality's OP/BE tilt is comparable in magnitude to that of value, so this lack of comovement change eliminates a generic channel through which stagnation shifted comovement for profitability-tilted portfolios. Were there such a generic channel, quality would display a similar shift. It is only value, with its current valuation based sort, that experiences a notable comovement shift, indicating that the channel for this phenomenon moves through the valuation sort.

The lack of comovement change in quality leaves its loading increase to be explained. Since loading is a function of comovement and relative volatility, unchanged comovement with increased loading implies higher relative volatility. This is reflected in the scale ratio of 1.43 reported in table 10, which shows that QMJ became 43% more volatile relative to RMW under stagnation. SMB volatility declines from 11.7% to

9.1% and CMA volatility declines from 8.0% to 5.5%, and quality also experienced higher loadings on these factors alongside little-changing partial correlations, demonstrating that the RMW rise is likely not specific to profitability under stagnation. It is with the HML factor (and less significantly, the market) where we observe genuine comovement change.

We predicted that since QMJ doesn't sort on price, its loading on profitability would not increase. We must reject this hypothesis; the loading does increase, but only because of decreasing volatility of profitability relative to quality. That the loading change was not accompanied by a comovement change specific to profitability implies that stagnation does impact quality's loading on profitability in a different way than that of value. We previously observed that the OP/BE tilt of quality expanded, which shows a change in the portfolio's composition, cutting against our hypothesis. However, the portfolio's change in composition under stagnation did not lead to tighter comovement between profitability and quality.

Table 9: Partial correlations of all factor portfolios

	Normal	Stagnation
Value (HML)	0.323	−0.151
Momentum (UMD)	0.060	−0.036
Quality (QMJ)	0.679	0.664

Table 10: QMJ's partial correlations and loading decompositions

	Partial correlation		Loading		Scale
	Normal	Stagnation	Normal	Stagnation	ratio
Market	−0.394	−0.543	−0.137	−0.198	1.046
SMB	−0.160	−0.213	−0.057	−0.119	1.575
HML	−0.573	−0.329	−0.398	−0.177	0.776
RMW	0.679	0.664	0.444	0.619	1.427
CMA	−0.215	−0.265	−0.155	−0.244	1.272

Table 11: UMD's partial correlations and loading decompositions

	Partial correlation		Loading		Scale
	Normal	Stagnation	Normal	Stagnation	ratio
Market	−0.146	−0.180	−0.244	−0.180	0.596
SMB	0.071	−0.199	0.129	−0.356	0.985
HML	−0.124	−0.411	−0.368	−0.735	0.603
RMW	0.060	−0.036	0.148	−0.082	0.901
CMA	0.067	0.028	0.246	0.079	0.777

Scale ratio is (σ_y/σ_x) under stagnation divided by (σ_y/σ_x) in the normal regime, where σ_y and σ_x are the residual standard deviations and the indicated factor after partialling out the remaining factors. The loading ratio equals the partial-correlation ratio multiplied by the scale ratio.

The correlation between HML and QMJ decreases from −0.42 in the normal rates environment to −0.62 under stagnation, which matches our final prediction that returns from these strategies should become more negatively correlated as value's profitability loading decreases. Furthermore, this correlation becomes more negative while the partial correlation of QMJ with HML becomes less negative. This shows that the further negative correlation arrives through shared factor exposures as predicted by our profitability channel, as value's loading on RMW changes from positive to negative, while quality's loading on RMW remains positive.

The changes in correlation for portfolio returns with momentum returns (from −0.07 to −0.58 for value and from 0.10 to 0.49 for quality) are not predicted by our hypotheses, but do not contradict them either. The value-momentum correlation results from momentum becoming more anti-value, seen in table 11 where UMD partial correlation with HML decreases from −0.12 to −0.41.

Table 12: Correlation Structure

	Normal	Stagnation
Value–Quality	−0.423	−0.620
Value–Momentum	−0.073	−0.581
Momentum–Quality	0.095	0.488
PC1 variance share	0.485	0.709

In our factor regressions, value loads very heavily on CMA (0.96), which fits the findings in Fama and French (2015, 11). Quality has a large positive loading on RMW in the normal regime, and it increases significantly under stagnation. Momentum's relatively low R^2 compared to the other portfolios is consistent with our hypothesis that momentum is disconnected from profitability and our other factor strategies. For most of the portfolios, we cannot make strong statements about market betas due to lack of statistical significance. For quality, the market beta decreases from −0.14 in the normal rate environment to −0.20. Our hypotheses do not provide a mechanism for quality becoming more defensive under stagnation, though it is plausible that it results from a widening in quality's low-beta tilt. Since this result is significant only at the 10% level, it is unclear how much it complicates our findings.

Table 13: Factor regressions, July 1990–November 2021

	HML	UMD	QMJ
Alpha	-0.003	0.009**	0.004***
	(0.002)	(0.004)	(0.001)
Market	0.035	-0.244	-0.137***
	(0.052)	(0.158)	(0.018)
SMB	0.242***	0.129	-0.057**
	(0.080)	(0.191)	(0.026)
RMW	0.349***	0.148	0.444***
	(0.114)	(0.275)	(0.036)
CMA	0.961***	0.246	-0.155***
	(0.111)	(0.474)	(0.054)
HML		-0.368	-0.398***
		(0.257)	(0.039)
Stagnation	-0.002	-0.005	0.000
	(0.004)	(0.005)	(0.001)
Market x Stag	0.049	0.064	-0.061*
	(0.077)	(0.182)	(0.032)
SMB x Stag	0.190	-0.485*	-0.062
	(0.168)	(0.248)	(0.051)
RMW x Stag	-0.592**	-0.230	0.175**
	(0.257)	(0.311)	(0.077)
CMA x Stag	-0.103	-0.167	-0.089
	(0.189)	(0.565)	(0.106)
HML x Stag		-0.367	0.221***
		(0.287)	(0.062)
Num.Obs.	358	358	358
R^2	0.471	0.168	0.715

Each column regresses the monthly long–short return of the indicated strategy on the Fama–French five factors, a secular-stagnation dummy (D), and its interactions. HML excludes its own factor. Newey–West standard errors (6 lags) in parentheses.
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

One possible limitation of our results relies on the guidelines firms use when accounting that leads to our book equity data. Intangible capital is not calculated as part of book equity, so changes in intangible intensiveness over time may push profitable firms into the growth leg of the value portfolio for accounting reasons, rather than economic reasons.

We look at returns in the rising-rate period to complete our picture. The 30 month window is too small to do proper inference, but the results contextualise our findings and provide direction for further research. In the rising period, the quality premium increases to 6.71%, while the value premium decreases to -9.43%. An appropriate out-of-sample test would be to determine whether the profitability loadings of the strategies revert to their pre-stagnation levels, but we lack the sample window to test this.

5 Conclusions

This paper is a beginning to the investigation of portfolio performance under secular stagnation. Our research determines that value's loading decreasing on profitability through the portfolio's tilt is consistent with discount rate compression and growth scarcity. We rule out two other possible mechanisms through which profitability can impact factor strategies. We demonstrate that there is not a general change in price sorting, as profitability does not impact the momentum portfolio, and that there is not a generic shift in how OP/BE-tilted portfolios load on RMW, as quality's tilt moves less than that of value with no change in comovement. We rejected the hypothesis that stagnation would not impact quality's OP/BE-tilt and profitability loading; instead we found that stagnation impacted quality less than value.

Future work should synthesize a tighter method for dating stagnation and non-stagnation periods so that the borders of the period can be made more concrete. Two findings of this paper: that quality may become more defensive with respect to market exposure under stagnation, and that quality becomes more correlated with momentum, should also be examined. Should the theory that structural factors depress the natural rate of interest and potential output growth hold, secular stagnation conditions may recur. The determinants of portfolio returns under such conditions need to be better understood given this potentiality.

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